

SATHWIK KARRA

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SUMMARY

Data Engineer with 3+ years of experience in designing, building, and maintaining scalable ETL/ELT pipelines and cloud-based data solutions. Proficient in Azure Databricks, PySpark, SQL, Azure Data Factory, Delta Lake, and Azure Storage. Experienced in implementing metadata-driven pipelines, Medallion architecture, SCD Type 2, data quality validation, and performance optimization for large-scale data processing. Strong understanding of data warehousing, data modelling, CI/CD, and Agile methodologies. Recognized for delivering reliable, high-quality data solutions that support business reporting and analytics.

SKILLS

Languages:	Python, SQL, PySpark
Big Data:	Apache Spark, Spark Streaming, Hadoop, Hive
Cloud:	Azure Databricks, Data Factory, Adls, BlobStorage
Data Engineering:	ETL/ELT, Medallion Architecture, DeltaLake, ScdType1, ScdType2, CDC
Data Warehouse:	Star schema & Snowflake Schema
DevOps & Version Control:	GitHub, CI/CD
BI& Reporting:	Power BI, DAX

PROFESSIONAL EXPERIENCE

Company: SPA LIMITED | assigned to AMEY LIMITED

May 2024 - present

Role: Data Engineer

Project description: Amey wanted a centralized system to manage its field operations. The company needed to track the real-time location of its vehicles, monitor employees' work status, and record issues reported at construction sites. Their manual process made it difficult to monitor project progress, allocate resources efficiently, and respond quickly to site issues. Our goal was to build a data platform that automated data collection, processing, and reporting for operational insights.

Tools used: Spark, PySpark, python, Azure Databricks, Databricks workflow, Databricks jobs, Git hub, CI/CD

Responsibilities:

- Designed and maintained a scalable metadata-driven ELT framework capable of supporting multiple data sources
- Implemented the Medallion Architecture (Raw, Bronze, Silver, and Gold) using Delta Lake to improve data quality, governance, and performance.
- Designed and managed Unity Catalog for centralized data governance, access control, and namespace management across environments.
- Developed reusable PySpark frameworks to ingest data from multiple REST APIs, supporting both full and incremental data loads.
- Processed nested JSON datasets using Spark DataFrame and flattened data using EXPLODE.
- Implemented data cleansing, validation, standardization, deduplication, and added audit columns (Created Date, Updated Date, Load Timestamp) and extracted relevant features from large datasets for downstream processing.

- Built incremental loading strategies using SCD Type 2 with Delta Lake Merge operations to maintain complete historical records.
- Automated workflow orchestration using Databricks Workflows and Jobs, including dependency management, scheduling, notifications, and monitoring.
- Integrated Git-based version control and CI/CD practices to streamline deployment across development, test, and production environments.

Company: SPA LIMITED | assigned to AND.E

Aug 2023-April 2024

Role: Data Engineer

Project description: The AND.E insurance company wanted to centralize policy, claims, and customer data from multiple systems. The goal was to build a scalable data platform that automated data ingestion, transformation, and reporting, providing accurate and timely insights for business operations and decision-making.

Tools used: AWS S3, Snowflake, Snowpipe, SQL

Responsibilities:

- Designed a batch ETL pipeline in Snowflake to load raw data from AWS S3 by configuring external stages and file formats.
- Designed Snowpipe to automate data ingestion, enabling files landing in S3 to be loaded into Snowflake without manual intervention.
- Implemented the COPY command for bulk data loading into raw staging tables.
- Implemented a layered architecture consisting of Raw, Staged, and Presentation layers for modular and maintainable data transformations.
- Handled duplicates, null values, data type mismatches, derived columns, and data normalization before promoting data to the next layer.
- Routed records that failed data quality validations to reject tables for further analysis and remediation.
- Loaded validated and cleansed data into the staged layer for downstream processing.
- Modelled sales and inventory data using a star schema with SCD Type 2, improving Power BI report performance by 25%.
- Implemented Snowflake Tasks to automate scheduled batch processing and data pipeline execution

Database Engineer (Internship) **Edinburgh Napier University**

Jan 2023-May 2023

- Designed, developed, and maintained SQL database to meet the requirements, ensuring data integrity and Performance.
- Created and managed database tables, including data insertion, updates, modifications, and retrieval
Using SQL queries (SELECT, JOINS etc..)
- Developed optimized SQL scripts for data extraction and transformation to support business Intelligence needs.
- Worked with a version control system managing repositories, branches, commits, pull requests, and merges for seamless collaboration.

EDUCATION

Master's in Computer Science

Edinburgh Napier University

Jan 2022- Jul 2023